

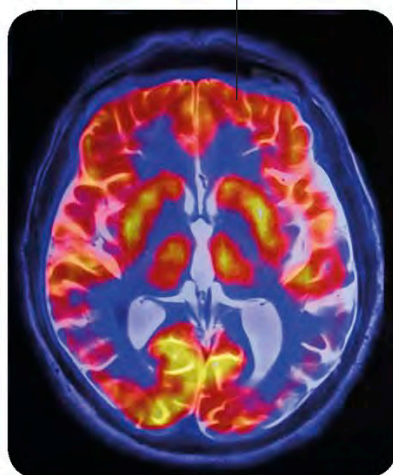
Uses



These lenses contain rubidium, which aids night vision.

Night-vision goggles

The structure of the brain can be seen clearly because of the use of radioactive rubidium.



PET scan

This sensitive device detects light by using a rubidium compound.

Photomultiplier



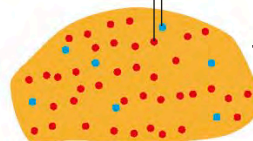
RUBIDIUM-STRONTIUM DATING

About a quarter of all rubidium atoms are radioactive. Slowly over time, they break down into strontium atoms. Comparing the amounts of these elements in a rock shows when that mineral was formed. Older rocks have less rubidium and more strontium in them.

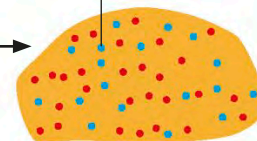
Rubidium-87 atoms (red) decay at a predictable rate.

Only small amounts of strontium-87 (blue) in the rock.

The amount of strontium-87 has increased over time.



Millions of years ago



Present day



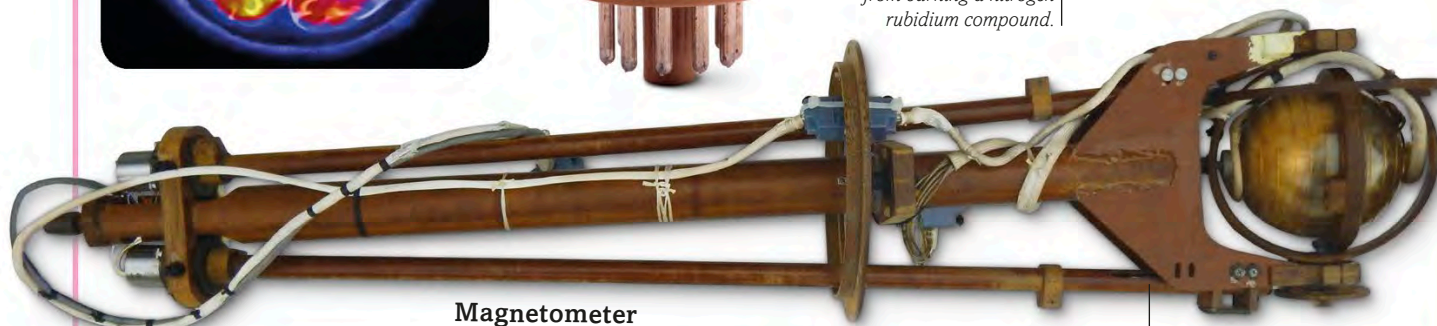
This purple colour comes from burning a nitrogen rubidium compound.

Fireworks

Electricity cables are hung from these rubidium-rich insulators.



Ceramic insulator



Magnetometer

This device from the early 20th century used rubidium to measure the strength of magnetic fields.

rubidium in it but is very rare. Rubidium atoms are sensitive to light and can be used in photoelectric cells (devices that convert light energy into electricity) and **night-vision equipment**. This element has radioactive forms, which can be used to measure the age of rocks. When injected into a

patient's body, rubidium targets tumours, which show up clearly on **PET (positron emission tomography) scans**. Rubidium is also used by light-sensitive electronics called **photomultipliers**, and in making **insulators** for high-voltage cables and some special types of glass.